Sex Determination

Worksheet · PU-II Biology, Chapter 4

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Worksheet – Sex Determination

15 questions · answer key on the last page

Q1. Who discovered the genetic/chromosomal basis of sex determination while studying insects, observing structures he named "X bodies"?

(a) Thomas Hunt Morgan
(b) Henking
(c) Walter Sutton
(d) Karl Landsteiner

Q2. The "X body" observed by Henking during spermatogenesis was later identified as:

(a) An autosome
(b) The X chromosome
(c) The Y chromosome
(d) A mitochondrion

Q3. Chromosomes directly involved in sex determination are collectively called:

(a) Autosomes
(b) Allosomes
(c) Alleles
(d) Loci

Q4. In humans, the sex chromosome constitution of a female is:

(a) XY
(b) XX
(c) XO
(d) ZW

Q5. In humans, the sex chromosome constitution of a male is:

(a) XX
(b) XY
(c) ZW
(d) ZZ

Q6. In the XO type of sex determination (e.g. grasshopper), the male has the constitution:

(a) XX
(b) XY
(c) XO (a single X, no Y)
(d) ZW

Q7. In the ZW type of sex determination, which sex is heterogametic?

(a) Male
(b) Female
(c) Both sexes equally
(d) Neither sex

Q8. In honey bees, unfertilised eggs develop, via parthenogenesis, into:

(a) Diploid females (queens or workers)
(b) Haploid males (drones)
(c) Sterile workers only
(d) Diploid males

Q9. In the XY type of sex determination in humans, all ova (egg cells) produced by the female carry:

(a) X chromosome only

(b) Y chromosome only
(c) Either X or Y in equal numbers
(d) No sex chromosome

Q10. In human males, sperm cells produced are:

(a) All X-bearing
(b) All Y-bearing
(c) 50% X-bearing and 50% Y-bearing
(d) 25% X-bearing and 75% Y-bearing

Q11. If an X-bearing sperm fertilises the egg in humans, the resulting zygote will develop into a:

(a) Male
(b) Female
(c) Either sex randomly
(d) Intersex individual

Q12. Which of the following organisms shows the XO type of sex determination?

(a) Humans
(b) Birds
(c) Grasshopper
(d) Drosophila

Q13. In the ZW system of sex determination, the male has the chromosome constitution:

(a) One Z and one W
(b) A pair of Z chromosomes (ZZ)
(c) A pair of W chromosomes (WW)
(d) XY

Q14. Which group of organisms is an example of the ZW (female heterogametic) system?

(a) Insects such as grasshoppers
(b) Mammals such as humans
(c) Birds
(d) Honey bees

Q15. "Male heterogamety" is best illustrated by which sex-determination systems?

(a) ZW only
(b) XO and XY
(c) Haplodiploidy only
(d) ZZ only

Answer Key & Explanations

Q1. (b) Henking — Henking discovered the chromosomal basis of sex determination in insects, observing specific nuclear structures during spermatogenesis that he named X bodies.

Q2. (b) The X chromosome — The X body Henking observed was later shown to be a chromosome and was named the X chromosome.

Q3. (b) Allosomes — Sex chromosomes, which are directly involved in sex determination, are collectively called allosomes.

Q4. (b) XX — Human females have the sex chromosome constitution XX, while males have XY.

Q5. (b) XY — Human males have the sex chromosome constitution XY, receiving an X from the mother and a Y from the father.

Q6. (c) XO (a single X, no Y) — In the XO system, there is no Y chromosome; males have only one X chromosome (XO) while females are XX.

Q7. (b) Female — In the ZW system, the female is heterogametic, having one Z and one W chromosome, while the male is homogametic (ZZ).

Q8. (b) Haploid males (drones) — In the haplodiploid system of honey bees, unfertilised eggs develop by parthenogenesis into haploid male drones.

Q9. (a) X chromosome only — All ova produced by human females carry an X chromosome, since females are XX.

Q10. (c) 50% X-bearing and 50% Y-bearing — Human males are XY, so they produce two types of sperm in equal proportion: 50% carrying X and 50% carrying Y.

Q11. (b) Female — Fertilisation by an X-bearing sperm produces an XX zygote, which develops into a female.

Q12. (c) Grasshopper — Grasshopper (along with earthworm) is an example organism showing the XO type of sex determination, where males lack a Y chromosome.

Q13. (b) A pair of Z chromosomes (ZZ) — In the ZW system, males are homogametic with a pair of Z chromosomes (ZZ), while females have one Z and one W.

Q14. (c) Birds — Birds, along with some reptiles and fishes, show the ZW type of sex determination, where the female is the heterogametic sex.

Q15. (b) XO and XY — Male heterogamety is seen in the XO and XY systems, where males produce two different types of gametes (with or without X, or with X or Y).